

Trident: Tactical Research, Inspection, and
Debris ENGagement Tool (UUV)

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A THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENT FOR THE DEGREE OF
BACHELOR OF ENGINEERING IN ROBOTICS AND AI ENGINEERING
SCHOOL OF ENGINEERING
KING MONGKUT'S INSTITUTE OF TECHNOLOGY LADKRABANG
2026

KMITL-2026-EN-X-XXX-XXX

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Thesis	Trident: Tactical Research, Inspection, and Debris ENgagement Tool (UUV)
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Degree	Bachelor of Engineering
Program	Robotics and AI Engineering
Year	2026
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ABSTRACT IN ENGLISH

Thailand's marine ecosystems are increasingly threatened by seabed pollution, habitat degradation, and insufficient infrastructure inspection due to the high cost and inaccessibility of traditional underwater monitoring systems. This project presents TRIDENT, a modular, low-cost Unmanned Underwater Vehicle (UUV) developed to support marine research, debris removal, and infrastructure maintenance in coastal regions. TRIDENT integrates a vision-based system for marine debris detection, a robotic arm module for interaction tasks, and high-resolution imaging for biodiversity documentation. Built using moldcast epoxy resin and silicone for cost-effective waterproofing, the vehicle incorporates an efficient ballast tank system and pressure sensors for quasi-static depth control. Realtime data acquisition is enabled via LAN communication with a Raspberry Pi and Arduino-based control architecture running ROS. With a compact and energy-efficient design, TRIDENT enhances underwater exploration in confined environments while addressing 11 critical marine challenges. The system contributes to sustainable ocean monitoring by reducing operational costs, improving inspection accessibility, and enabling data-driven conservation efforts.

ACKNOWLEDGEMENT

We would like to sincerely thank Asst. Prof. Dr. Jutarut Chaoraingern for her valuable guidance and continuous support throughout this project. Her insights were essential to the success of TRIDENT.

We also thank the School of Engineering, KMITL, for providing the knowledge and resources that enabled us to carry out this work. The skills gained from various courses

greatly contributed to our design and implementation. Lastly, we appreciate the support from our families, friends, and teammates for their encouragement and collaboration during this journey.

Nabhatara Tongsri

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LIST OF ABBREVIATIONS AND SYMBOLS

UUV	Unmanned Underwater Vehicle
ROV	Remotely Operated Vehicle
AUV	Autonomous Underwater Vehicle
BLDC	Brushless DC Motor
OD	Outer Diameter
ID	Inner Diameter
M3	Metric 3mm Lead Screw
M4	Metric 4mm Lead Screw
M6	Metric 6mm Lead Screw

CHAPTER 1

INTRODUCTION

1.1 SIGNIFICANCE AND BACKGROUND OF THE PROBLEM

1.1.1 Noise Pollution in UUV

Traditional UUV came packaged with noise pollution, whether it being from its motors, acoustic sensors, or even water disperse around its shape. This toughens the recording and tracking attempt for various marine species.

Big commercial submarines adopt another method providing more control, and almost no noise is created. A ballast tank, which is by all mean a better method for a lengthy period hovering, compared to other UUV. But this is hard to implement in a small-scale submersible.

1.1.2 Enhancing Marine Species Monitoring in Thailand

Monitoring pollution levels and marine biodiversity is crucial for managing ecosystems sustainably. As markers of the health of the environment, marine species' loss frequently indicates more serious problems with habitat degradation, temperature changes, or water quality.

However, traditional maritime monitoring techniques that are commonly used in Thailand, such as diver-based surveys or equipment installed on boats, are risky, time-consuming, and labor-intensive. These techniques are further constrained by depth limitations, strong currents, and low vision.

1.1.3 Ensuring Maintenance of Underwater Systems

Underwater systems, for example, pipelines, fish farms, and marine research equipment, are constantly exposed to harsh environmental conditions including corrosion, and damage from marine life. These systems require regular inspection, repair, and maintenance to remain functional and safe.

Traditional inspection methods rely on divers or expensive Remotely Operated Vehicles (ROVs), which are costly, time-consuming, and pose safety risks especially in deep or murky waters. As a result, many installations suffer from delayed maintenance and undetected failures, leading to higher repair costs or system downtime.

1.1.4 Traditional UUV is costly

Unmanned Underwater Vehicles are essential for tasks such as marine research, environmental monitoring, and infrastructure inspection. However, most traditional UUVs are expensive and inaccessible to small research labs, universities, and local conservation groups. For instance, Chasing gladius Mini (35,000 –60,000 Thai baht), BlueROV2 (180,000 Thai baht), and VideoRay Pro 5 (500,000 – 3,700,000 Thai baht). These systems also require additional costs for operation, deployment, and training. Due to these barriers, many underwater systems go uninspected, and marine research is limited—especially in developing regions like Thailand.

1.2 OBJECTIVES OF THE PROJECT

The primary objective of this project is to design and develop multipurpose Unmanned Underwater Vehicle (UUV) that supports various underwater applications in both research and maintenance. The system is aimed to provide affordable, modular, and adaptable platform that can operate effectively in canal and coastal environments of Thailand.

Key Objectives

1. Develop a Modular Multi-Use UUV Platform:

Create a compact and cost-effective UUV structure that allows easy integration of various tools and sensors based on mission requirements. Design the system to support plug-and-play modules for quick reconfiguration.

2. Enable Underwater Debris Detection:

Integrate vision-based detection for identifying marine litter and seabed debris. Attach basic cleaning tools or manipulators for collecting or tagging debris for environmental maintenance.

3. Support Marine Biological Research (Ichthyology):

Equip the UUV with high-resolution cameras to monitor marine life habitats and behaviors. Enable data collection for research in ichthyology and marine biodiversity mapping.

4. Monitor and Inspect Underwater Infrastructure:

Facilitate routine inspection of underwater systems such as pipelines, fish cages, research sensors, and marine installations. Provide real-time visual data for structural assessment and potential failure detection.

5. Reducing Noise Pollution in UUV:

Reduce the noise pollution created by other modern UUV, by reverting to a traditional mean. A fully controllable ballast tank and maneuverable hovering.

1.3 SCOPE AND LIMITATIONS OF PROJECT

To minimize manufacturing costs while maintaining structural functionality, the connecting components of the Unmanned Underwater Vehicle (UUV) were fabricated by 3D-printing PETG components before applying a resin shell around. This method allows for rapid prototyping, ease of customization, and acceptable water resistance, particularly in shallow-to-moderate depth applications. Traditional 3D-printed materials such as PLA, while accessible, exhibit poor waterproofing and dimensional stability in underwater environments, making them unsuitable for critical structural components.

Despite its advantages, PETG with resin coat exhibits lower mechanical properties. Specifically, tensile strength, impact resistance, and fatigue tolerance when compared to industrial-grade polymers (e.g., polycarbonate, PEEK) or metallic materials. These limitations must be considered in operational contexts involving hydrostatic pressures, mechanical vibrations, or potential collision events. Finite Element Analysis (FEA) is recommended for evaluating stress concentrations within these components under load conditions.

Given the spatial constraints inherent in the UUV's compact form factor, the onboard battery modules are limited in both physical size and capacity, leading to a reduction in operational endurance. This imposes constraints on mission duration and necessitates efficient power management strategies, such as pulse-width modulation (PWM) for thruster control and low-power sleep modes for auxiliary systems. Trade-offs between

vehicle autonomy and power density are actively evaluated.

For underwater communication, Bluetooth and other RF-based protocols are infeasible due to rapid signal attenuation in conductive media like seawater. To overcome this, the UUV employs a tethered Ethernet-based Local Area Network (LAN) for high-throughput, low-latency communication during deployment and testing. This configuration supports real-time control, telemetry acquisition, and bulk data transfer, which are critical for debugging, mission uploads, and sensor data retrieval.

The vehicle is equipped with two ballast tanks designed to modulate its net buoyancy, allowing for depth control without constant thrust input. By achieving near-neutral buoyancy and increasing vertical displacement, the UUV can maintain submerged states for extended periods. This feature is advantageous for the UUV's primary objective: Reducing Noise Pollution in UUV. Reduced propulsion noise and acoustic signal created by other UUV.

1.4 Methodology Used in the Project

TRIDENT is designed to have a highly compact form factor, as its structure contains air, which increases buoyancy and makes submersion more challenging due to density differences. Since one of the primary objectives of the project is to navigate confined spaces, minimizing the vehicle's size enhances its ability to operate efficiently in such environments.

1.4.1 Software Development

The robot primarily operates using ROS (Robot Operating System), which provides a flexible framework for handling navigation, sensor integration, and autonomous decision-making. For underwater mapping, the robot employs Simultaneous Localization and Mapping (SLAM), a technique that allows it to construct a map of the seabed while simultaneously determining its own position within that environment. To achieve this, the system utilizes laser distance sensors, which play a crucial role in detecting the contours and features of the seabed in real time. These sensors, specifically infrared (IR) distance sensors from Sharp, are designed to measure distances within a range of 20 cm to 150 cm. They operate based on triangulation and reflected infrared light intensity, where the emitted IR beam is reflected off surfaces and detected by an internal position-sensitive detector (PSD). This data is then processed to estimate distances accurately, enabling the robot to create a detailed and precise underwater map while navigating autonomously.

1.4.2 Communications

The TRIDENT robot utilizes LAN communication for underwater operations, as water can completely block the radio frequency signals required for other forms of communication. Within the robot, a Raspberry Pi serves as the primary control unit, running ROS (Robot Operating System) to manage and facilitate communication with an Arduino Nano. The Arduino Nano is responsible for controlling the sensors and motors via UART (Universal Asynchronous Receiver-Transmitter). UART is a serial communication protocol that enables data transfer between devices without the need for a shared clock signal. Data is transmitted one bit at a time, including a start bit, data bits, an optional parity bit, and a stop bit. This protocol is commonly used in embedded systems, microcontrollers, and computers, supporting full-duplex communication through transmit (TX) and receive (RX) lines.

1.5 Project Implementation Plan

1.5.1 Independent Site Selection and Mission Planning

Site selection will be conducted independently based on publicly available data, prior studies, and observed environmental needs. Emphasis will be placed on identifying locations suitable for underwater inspection, ecological observation, and experimental data collection.

1.5.2 Data acquisition and analysis

The UUV will be deployed to collect data on Marine species, and underwater infrastructure. The collected data will be analyzed to assess the health of marine ecosystems, and evaluate the condition of underwater infrastructure.

1.5.3 Marine Species Documentation and Monitoring

The UUV will be employed to capture video and image data of marine species within their natural habitats. Which crucial for biodiversity assessments, species identification, and behavioral studies, contributing to a deeper understanding of the local marine ecosystems.

1.5.4 Underwater Infrastructure Inspection

Routine inspections of underwater infrastructure, including pipelines, fish cages, and other installations. These inspections will facilitate the detection of structural damage, corrosion, and other potential issues, enabling timely maintenance and preventing costly failures.

1.6 Benefits Derived from the Research

1.6.1 Data-Driven Conservation

The UUV will provide data that supports informed decision-making in marine conservation and management. This data will enable targeted interventions and enhance the effectiveness of conservation efforts.

1.6.2 Improved Underwater Infrastructure Maintenance

By deploying the UUV for routine inspections of underwater infrastructure, such as pipelines and fish cages, the project will reduce the reliance on expensive and potentially hazardous diver-based inspections. Increasing driver safety.

1.6.3 Enhanced Research Capabilities

The UUV will enable researchers to access and study underwater environments that are otherwise difficult to reach, facilitating new discoveries and advancing scientific understanding of marine ecosystems.

1.7 Capstone Design Details

1.7.1 Application of Knowledge and Skills from Previous Courses

Subject that lie closely to the project include

01006723: General Physics 1 classical mechanics, including motion, forces, energy, and buoyancy

01006725: General Physics 2 electronic circuits, understanding motor behavior, battery systems

01416312: Electric Circuit and Electronics designing and building the UUV's electronic systems, including power distribution, sensor connections, motor controllers

01416305: Artificial Intelligence Technology vision-based object detection

01416308: Kinematics and Dynamics analyzing forces and torques on the body, predicting behavior under water currents, and developing motion control algorithms

01416500: Computer Vision image processing, object detection, and navigation using onboard cameras

01416306: Safety and Standardization in RAI compliance with underwater safety standards (IP ratings, battery safety)

01416319: Manufacturing Process Applied in the mold-based component fabrication using epoxy resin and silicone, and in selecting suitable materials and production methods for waterproof, strong, and modular design

01416511: Mobile Robots autonomous navigation, localization, path planning, sensor fusion, and the integration of mechanical and software systems in mobile robotic platforms

01416408: COM AID Mechanical Design CAD modeling, stress analysis, and mechanical design optimization for components like the ballast tank, motor housing, and main frame structure

1.7.2 Design of a Device

For a modular use UUV platform, the robot must be compatible with a tool such as robot arm for collecting debris or holding on to pipeline. For inspection and monitoring, the device is equipped with a camera module, two hydrostatic pressure sensors and two laser distant sensors. These sensors enable the use of Point Cloud for mapping the underwater bed for more data collecting.

Additionally, focusing primarily on minimizing environmental impact, the design of our robot is centered around this concept. Two key improvements over traditional UUVs are (1) the elimination of noise pollution generated by the motor and (2) reducing the cost of the robot by optimizing its design. Additionally, the robot is designed to have a smooth, rounded shape with as few openings as possible to better withstand the challenges of Thailand's polluted waterways.

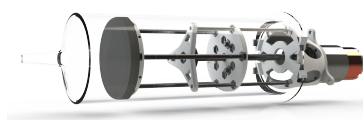


Figure 1.1 Ballast Tank Design

As shown in Figure 1.1, due to its large size and integration challenges, the Ballast Tank is not ideal for small UUV applications. However, it eliminates the need for multiple motors in depth control, thereby

reducing noise and disturbances.

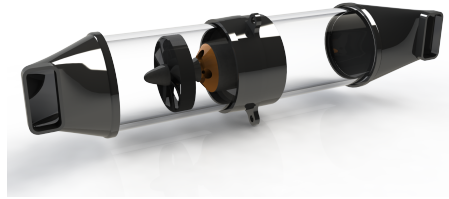


Figure 1.2 Motor Tube Design

As shown in Figure 1.2, inspired by the mechanism of fish gills, two brushless motors are enclosed within the robot's structure to achieve two primary objectives: (1) minimizing noise pollution and water disturbances by preventing direct motor contact with the external environment, and (2) shielding the motors from debris commonly found in Thailand's polluted waterways.

Table 1.1 Power Consumption

Components	Amount(Pieces)	Voltage (V)	Current (A)	Total Current (A)
Raspberry Pi	1	5.1	2.5	2.5
Arduino Nano	1	6-12	0.03	0.03
Voltage Sensor	1	11.1	0.05	0.05
Pressure Sensor	1	5	0.03	0.03
GY-86 IMU/10DOF	1	5	0.02	0.02
BLDC A2212 930kV	2	10	2	4
Required		11.1		6.63

Using 11.1V Lipo battery, 2200mAh, the robot can operate for approximately 20 minutes. This is calculated by dividing the battery capacity (2.2 Ah) by the total current draw (6.63 A), resulting in an estimated runtime of around 0.33 hours or 20 minutes under continuous operation at full load. However, actual runtime may vary based on factors such as such motor usage patterns, sensor activity, and power management strategies implemented in the robot's design.

Moreover a 11.1V Lipo battery, 3000mAh, is also available, which can provide approximately 27 minutes of operation under the same conditions. This is calculated by dividing the battery capacity (3 Ah) by the total current draw (6.63 A), resulting in an estimated runtime of around 0.45 hours or 27 minutes under continuous operation at full load.

The ballast tank battery is separated from the main battery, hence creating a redundant powersystem. It consists of a 7.4V Lipo battery at 1.1Ah. This act as a fail-safe mechanism, ensuring that the ballast tank can operate independently of the main system. This design choice enhances the reliability and safety of the UUV, allowing it to maintain buoyancy control even if the main battery is depleted or encounters issues during operation.

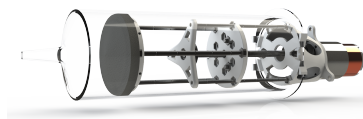


Figure 1.3 System Architecture Diagram

The UUV works by using sensors to understand its position and surroundings. First, an IMU (Inertial Measurement Unit) is used to measure the UUV's movement and direction, while a laser sensor detects nearby walls and obstacles also pressure sensor uses to sense the pressure under water giving how depth the robot is. These sensors are connected to an Arduino, which reads their data and sends it to a Raspberry Pi. The Raspberry Pi acts as the brain of the system, processing the sensor data using ROS (Robot Operating System). By combining IMU and laser sensor readings, the Raspberry Pi calculates the UUV's exact position underwater and detects its surroundings. It then adjusts the thrusters to keep the UUV stable and on course. The UUV communicate directly from the computer using LAN cable.

For a fail-safe system, we connect the POP32i module to two DC motors, which operate using a backup power supply (7.4V, 9600 mAh). The system monitors whether the Arduino Nano is functioning. If the Arduino Nano fails to operate within a certain time, the POP32i module will activate the fail-safe DC motor, causing the UUV to float back to the water surface.

1.7.3 Compliance with Industrial Design Standards, Codes, and/or Specifications

ISO 13628-8: This standard provides guidelines for remotely operated tools and vehicles for subsea use. It focuses on functionality, safety, and interoperability.

ISO 19007: For materials used in underwater vehicles, focusing on testing for microplastic and corrosion resistance.

IEC 60533: Electromagnetic compatibility for electrical systems in ships and underwater vehicles.

IP Ratings (Ingress Protection): Ensure electrical systems are rated for water ingress, e.g., IP68 or higher for components exposed to high pressures.

NFPA 70 (NEC): Ensures electrical wiring and installations meet underwater safety standards.

IEC 62619: For rechargeable lithium batteries in industrial applications, including underwater systems.

UL 1642: For lithium-ion battery safety, ensuring thermal runaway, short circuits, and water ingress protection are considered.

1.7.4 Incorporation of Realistic Technical and Non-Technical Design Constraints

- **Technical Constraints**
 - Size and weight limitations to ensure neutral buoyancy and compact form factor
 - Waterproofing requirements for all submerged components
 - Space constraints for component integration inside acrylic tubes
 - Power management and battery life optimization
 - Sensor and controller compatibility with Raspberry Pi and Arduino Nano
 - Manual manufacturability using 3D printing and resin coating
- **Non-Technical Constraints**
 - Safety precautions for high-speed propellers, electrical wiring, and sealing
 - Availability of parts within project timeline and supplier delivery constraints
 - Ease of maintenance and reusability of system components for educational use
 - Budget constraints for materials, components, and manufacturing processes
 - Environmental considerations for materials used in aquatic environments

1.7.5 Teamwork, Report Preparation, and Oral Presentation

Team's Management and Collaboration Strategies

The allocation of tasks in this project is based on the specialized skills and expertise of each team member to ensure maximum accuracy and efficiency. Responsibilities are clearly defined, along with specific timelines for each task.

Given the need for precision and meticulous attention to detail, effective communication and collaboration play a crucial role throughout the project. Continuous coordination is essential to maintaining the accuracy and consistency required for successful completion.

Report Preparation

Regarding the report preparation, the group regularly discusses and consults to update progress and determine work strategies. Weekly meetings ensure that all members have a fundamental understanding of the topics and content required for the report. However, further discussions are necessary to refine the writing and ensure clarity and relevance. Additionally, continuous review of both content and formatting is conducted to minimize redundancy and enhance overall coherence.

Plan for Presentation

The presentation slides have been developed and initially presented to the project advisor, as well as to senior students for feedback. This experience has provided the team with valuable insights; however, due to the high-pressure nature of the final presentation, meticulous preparation is crucial. Efforts are being made to refine the content further, ensuring a thorough understanding of all aspects of the project.

CHAPTER 2

RELATED THEORIES AND LITERATURE REVIEW

- 2.1 RELATED THEORIES
- 2.2 LITERATURE REVIEW
- 2.3 SUMMARY

CHAPTER 3

DESIGN AND CONSTRUCTION

3.1 Mechanical Design

3.1.1 Ballast Tank System

The ballast tank (Figure 1.1) utilizes a 500ml syringe for ease of manufacturing and control. An M6 lead screw guides a threaded coupling along the rod, which is coupled to a JGB37-520 motor at the terminal. The ballast tank is divided into two main sections: the mobile section and the stationary section.

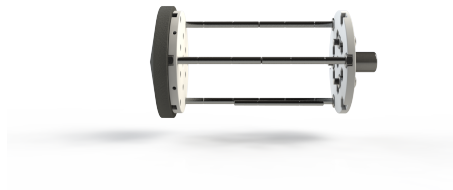


Figure 3.1 Mobile Section of Ballast Tank

The mobile section (Figure 3.1) consists of two 3D-printed parts connected by three M4 lead screws, each 100mm in length. The front-facing part of the syringe has a rubber seal, while the aft section features a threaded coupling. This section can move independently along the syringe's radial axis.

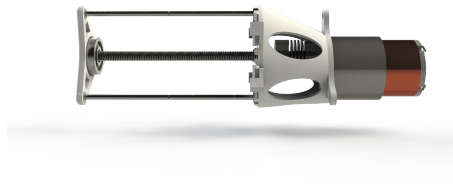


Figure 3.2 Stationary Section of Ballast Tank

The stationary section (Figure 3.2) is the main syringe body, with its aft end joined to a 3D-printed cap with a snap-fit mechanism. This cap connects to the motor and houses the motor-M6 coupler. Three M3 lead screws are bonded to the end cap, acting as guiding rails for the mobile section. At the tip of each screw, a 3D-printed cap holds them perpendicular, and a bearing is attached to ensure smooth rotation of the main M6 lead screw.

3.1.2 Motor Tube Compartment

Figure 1.2, the propulsion system utilizes A2212 930kV brushless DC (BLDC) motors, each securely mounted onto a coated structural component positioned between two acrylic tubes. These tubes have an outer diameter (OD) of 50 mm, a thickness of 3 mm, and a length of 70 mm. At the end of each tube, an outlet facilitates controlled water intake and expulsion, enabling efficient fluid flow management within the system.



Figure 3.3 Propellor Design

Figure 3.3 illustrates the propeller design, which is a 3D-printed part coated with resin layer. The propeller is designed to have 8 blades. The blades are angled to optimize thrust while minimizing drag, ensuring efficient propulsion for the underwater vehicle.

The propeller also have a circular ring around the blades, which serves to enhance the structural integrity of the propeller . The ring helps to stabilize the flow of water around the blades, creating a streamline laminar output flow and reducing turbulence, which can improve the overall noise performance of the propulsion system.

3.1.3 Circuit Compartment

The circuit tube is designed to be partially sealed, leaving only the necessary space for cable connections. This design ensures a partial safety precaution against small leaks, with all core components securely fitted inside to prevent interference with other parts of the robot.

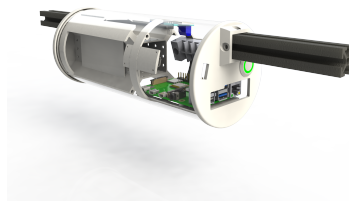


Figure 3.4 Circuit Compartment Design

Figure 3.4 shows the circuit compartment design. The circuit is housed within an acrylic tube with an outer diameter (OD) of 90 mm, a wall thickness of 3 mm, and a length of 220 mm. At each end of the tube,

a 3D-printed cap is securely snapped into place, enclosing the compartment while also serving as a connector for a 100 mm-long 1010 aluminum profile. These aluminum profiles function as mounting structures for the sensor system, providing stability and ease of integration.

Enclosed within the acrylic tube is the primary control and power management system, consisting of a Raspberry Pi, an Arduino Nano, voltage sensors, an inertial measurement unit (IMU), a terminal, and a Lipo Battery for rechargeability and stability. These components collectively form the core circuitry housed within the enclosure.



Figure 3.5 Circuit Compartment Design Side View

To facilitate maintenance, all components are mounted on an lead 3D-printed rod, enabling a controlled sliding motion for easy access and adjustments.

Externally, the system integrates a laser distance sensor, hydrostatic pressure sensors, and a Raspberry Pi camera, enabling environmental perception and data acquisition for autonomous operation.

3.1.4 Main Frame Design

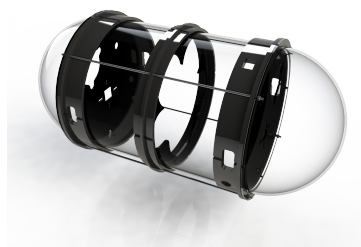


Figure 3.6 Main Frame Design

The robot's shell is divided into four distinct sections, each separated by an acrylic plane: the front dome, which encapsulates the Raspberry Pi Camera for visual data acquisition and the fail-safe module to ensure system redundancy; two central compartments designed to house the ballast tank, propulsion motor, and circuit enclosure; and the aft dome.

This modular design, with acrylic planes separating the compartments, optimizes component placement, provides easy access for maintenance and system integration, and ensures structural integrity and functionality. The central coupler is coated and bonded to two acrylic tubes with an outer diameter (OD) of 220 mm, a wall thickness of 3 mm, and a length of 10 mm. At the end of each tube, an additional coated coupler connects the dome to the tube, ensuring structural integrity.

Four M3 lead screws run along the sides of the robot, providing compression to maintain structural stability. The dome is designed for removal, as it is not permanently sealed. Instead, an O-ring (or a Rubber Seal), in combination with the applied compression, ensures a watertight seal while allowing for easy access to internal components.

3.2 Electrical Design

3.3 Control System Design

3.4 AI Detection System Design

3.5 Manufacturing Process

CHAPTER 4

EXPERIMENTAL RESULTS

4.1 EXPERIMENTAL SETUP

4.2 RESULTS

4.2.1 Result 1

4.2.2 Result 2

4.3 DISCUSSION

CHAPTER 5

CONCLUSION AND FUTURE WORK

- 5.1 CONCLUSION
- 5.2 CONTRIBUTIONS
- 5.3 FUTURE WORK

[?]

Appendix A
ADDITIONAL MATERIALS

AUTHOR BIOGRAPHY



Name Mr. First Name Last Name
Date of Birth January 1, 1990, in Bangkok